Linear Algebra 1

Adar's Study Series

LINEAR

ALGEBRA 1

Systems of Linear Equations · Echelon Forms and General Solutions
Vectors, Linear Combinations and Span · The Matrix Equation

A First Course in Linear Algebra

Deep Violet & Amber Edition

Adar's Study Series
Linear Algebra 1

Preface

This book is a first course in linear algebra. It begins where the subject begins — with systems of linear equations and the row operations that solve them — and develops, step by step, the central ideas of a first semester: echelon and reduced row echelon forms, pivots and free variables, vectors and their geometry, linear combinations and span, and finally the matrix equation $A\vec{x} = \vec{b}$ together with the structure of solution sets.

How the book is organised.

- **Chapter 1** introduces linear systems, the augmented matrix, elementary row operations, and the goal of row reduction.
- **Chapter 2** develops echelon form (REF) and reduced row echelon form (RREF), pivots and free variables, a general two-pass reduction method, and the three consistency cases.
- **Chapter 3** turns to vectors: the spaces $\mathbb{R}^n$, vector operations and their geometry, linear combinations, span, and the equivalence between systems, vector equations and span membership.
- **Chapter 4** studies the matrix equation $A\vec{x} = \vec{b}$: how $A\vec{x}$ is computed, and how a system is written in vector and matrix form.
- **Chapter 5** classifies solution sets: homogeneous and nonhomogeneous systems, the solution structure theorem, and the geometric picture in $\mathbb{R}^3$.

How to use this book. Every chapter contains worked examples carried out in full, highlighted answers, remarks that flag common pitfalls, and a summary of key points. The reader should work through the examples with pencil and paper; row reduction is learned by doing. No background beyond school algebra is assumed.

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Notation

- $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^n$, $\mathbb{R}^m$ — the real numbers and their n-fold products.
- Vectors are written with arrows: $\vec{x}$, $\vec{b}$, $\vec{v}_1$, $\vec{0}$.
- $[\,A \mid \vec{b}\,]$ — the augmented matrix of a linear system; the vertical bar separates coefficients from constants.
- Elementary row operations: $R_i \to kR_i$ (scaling), $R_j \to R_j \pm kR_i$ (replacement), $R_i \leftrightarrow R_j$ (interchange).
- In pattern matrices: $\blacksquare$ denotes any non-zero entry, $*$ any number, and 0 a zero entry.
- $\mathrm{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$ — the set of all linear combinations of the given vectors.
- $\Rightarrow$, $\Leftrightarrow$ — implies; equivalent to.

Contents

Chapter 1

Linear Equations and Systems

1.1. What Is Linear Algebra?

Linear Algebra is the branch of mathematics that studies linear equations, systems of linear equations, vectors, matrices, and linear transformations.

Main Goal

Find the values of unknown variables that satisfy one or more linear equations simultaneously.

Applications

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Data Science
- Computer Graphics
- Computer Vision
- Signal Processing
- Engineering

1.2. Systems of Linear Equations

Definition 1.1 (System of linear equations)**.** A **system of linear equations** is a collection of two or more linear equations involving the same variables.

General Form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

1.3. Solving by Hand: A System in Two Variables

Example 1.1 (Substitution and elimination)**.** Find x and y such that:

$$\begin{aligned} x - y &= -2 \\ 3x - 6y &= -15 \end{aligned}$$

Method 1 — Substitution

From equation (1):

$$x - y = -2 \quad \Rightarrow \quad x = y - 2.$$

Substitute into equation (2):

$$3(y-2) - 6y = -15 \quad \Rightarrow \quad 3y - 6 - 6y = -15 \quad \Rightarrow \quad -3y = -9 \quad \Rightarrow \quad y = 3.$$

Then:

$$x = y - 2 = 3 - 2 = 1.$$

Answer: (x, y) = (1, 3)

Remark 1.1. Limitation: the substitution method becomes messy as the number of variables and equations grows.

Method 2 — Elimination

Multiply equation (1) by 3:

$$3x - 3y = -6.$$

Subtract from equation (2):

$$(3x - 6y) - (3x - 3y) = -15 - (-6) \quad \Rightarrow \quad -3y = -9 \quad \Rightarrow \quad y = 3.$$

Substitute back into equation (1):

$$x - 3 = -2 \quad \Rightarrow \quad x = 1.$$

Answer: (x, y) = (1, 3)

Geometric Interpretation

Each equation represents a straight line in the 2D plane.

- Intersecting lines $\rightarrow$ One unique solution
- Parallel lines $\rightarrow$ No solution
- Same line $\rightarrow$ Infinitely many solutions

For Example 1.1, the two lines intersect at exactly one point:

Intersection Point: (1, 3)

1.4. Three Variables: A System with Infinitely Many Solutions

Example 1.2. Find x, y and z such that:

$$\begin{aligned} x - 2y + 4z &= 1 \\ x + y + 7z &= 3 \end{aligned}$$

Solving the System

Subtract equation (1) from equation (2):

$$(x+y+7z)-(x-2y+4z)=3-1 \quad\Rightarrow\quad 3y+3z=2 \quad\Rightarrow\quad y=\frac{2}{3}-z.$$

Substitute into equation (1):

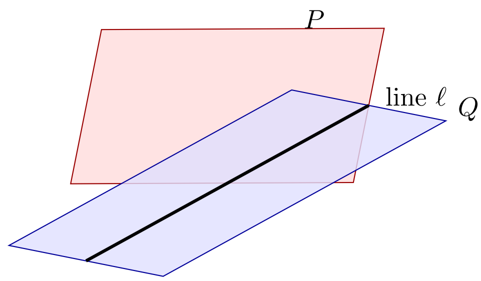
$$x-2\left(\tfrac{2}{3}-z\right)+4z=1 \quad\Rightarrow\quad x=1+\tfrac{4}{3}-2z-4z \quad\Rightarrow\quad x=\frac{7}{3}-6z.$$

Let the free variable $z=t$ (t is a parameter):

$$x=\frac{7}{3}-6t, \qquad y=\frac{2}{3}-t \quad \textbf{and} \quad z=t \qquad (t\in\mathbb{R})$$

Geometric Interpretation

Each equation represents a plane in 3-dimensional space. Two (non-parallel) planes generally intersect in a straight line — not a single point.



Two non-parallel planes P and Q intersect in a straight line ℓ.

Conclusion. This system has infinitely many solutions, forming a line parameterised by t.

1.5. Linear Equations in n Variables

General Form

$$a_1x_1+a_2x_2+\cdots+a_nx_n=b$$

Where:

- $x_1,x_2,\ldots,x_n \;\rightarrow\;$ Variables
- $a_1,a_2,\ldots,a_n \;\rightarrow\;$ Coefficients
- $b \;\rightarrow\;$ Constant

General Linear System

$$\begin{aligned}
a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n&=b_1\\
a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n&=b_2\\
&\;\;\vdots\\
a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n&=b_m
\end{aligned}$$

1.6. The Augmented Matrix

Definition 1.2 (Augmented matrix)**.** The **augmented matrix** packages every coefficient and constant of a linear system into a single array, separating the coefficients from the constants with a vertical bar:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

- **Left part** $\rightarrow$ Coefficient Matrix
- **Right part** $\rightarrow$ Constant Column

Goal of Row Reduction

Reduce the augmented matrix using elementary row operations until the solution of the system can be read off directly.

1.7. Elementary Row Operations

1. **Row Scaling:** $R_i \rightarrow kR_i \quad (k \neq 0)$. Multiply every entry of a row by a non-zero constant.
2. **Row Replacement:** $R_j \rightarrow R_j \pm kR_i$. Add/subtract a multiple of one row to another row.
3. **Row Interchange:** $R_i \leftrightarrow R_j$. Swap the positions of two rows.

Key Fact. Elementary row operations never change the solution set of the system — they only change its appearance.

1.8. Row Reduction in Action

Example 1.1 revisited via the augmented matrix

Starting augmented matrix:

$$\left[\begin{array}{cc|c} 1 & -1 & -2 \\ 3 & -6 & -15 \end{array}\right]$$

Apply the operation $-3R_1 + R_2 \rightarrow R_2$:

$$\left[\begin{array}{cc|c} 1 & -1 & -2 \\ 0 & -3 & -9 \end{array}\right]$$

From Row 2: $-3y = -9 \Rightarrow y = 3$. From Row 1: $x - y = -2 \Rightarrow x = 1$.

Answer: (x, y) = (1, 3)

Example 1.2 revisited via the augmented matrix

Starting augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 1 & 1 & 7 & 3 \end{array}\right]$$

Apply the operation $R_2 - R_1 \to R_2$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 3 & 3 & 2 \end{array}\right]$$

From Row 2: $3y + 3z = 2 \;\Rightarrow\; y = \frac{2}{3} - z$. Substitute the expression of y obtained from Row 2 into Row 1:

$$x = \frac{7}{3} - 6z.$$

Let $z = t$ (free parameter):

$$x = \frac{7}{3} - 6t, \qquad y = \frac{2}{3} - t \quad \textbf{and} \quad z = t \qquad (t \in \mathbb{R})$$

1.9. Looking Ahead: Echelon Form

Learning Goals

- Learn Echelon Form (EF) and Reduced Row Echelon Form (RREF).
- Solve linear systems using systematic Row Reduction.
- Determine the number of solutions of a linear system.

Types of Solutions

Inconsistent System

- No Solution

Consistent System

- One Unique Solution
- Infinitely Many Solutions

1.10. Summary

- Linear Algebra studies linear equations and systems of linear equations.
- An augmented matrix packages coefficients and constants into one array.
- Row reduction simplifies a system without changing its solution set.
- Three elementary row operations: Row Scaling, Row Replacement, Row Interchange.

- Echelon Form (EF) and Reduced Row Echelon Form (RREF) are standard forms used to solve systems efficiently.
- A linear system may have: No Solution, One Unique Solution, or Infinitely Many Solutions.

With row reduction in hand, the next chapter develops echelon form and reduced row echelon form in depth.

Chapter 2

Echelon Forms and General Solutions

2.1. Echelon Form

Definition 2.1 (Echelon form)**.** A matrix is in **echelon form** if it satisfies the following two conditions:

- All non-zero rows are above any **zero rows**.
- Each non-zero **leading entry** (the first non-zero number from the left in a row) is to the **left** of the non-zero leading entries of the rows below it.

Notation Key

- $\blacksquare$ or $1 \;=\;$ any non-zero number
- $* \;=\;$ any number
- $0 \;=\;$ zero entry

Examples of Echelon Form

A 2×3 example:

$$\begin{bmatrix} \blacksquare & * & * \\ 0 & 0 & \blacksquare \end{bmatrix} \qquad \text{A } 4 \times 6 \text{ example:} \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2.2. Reduced Row Echelon Form

Definition 2.2 (Reduced row echelon form)**.** A matrix is in **reduced row echelon form** if it meets the criteria for echelon form (Definition 2.1), and additionally satisfies:

- Every non-zero leading entry (**pivot**) equals 1.
- There are zeros above each non-zero leading entry (pivot), as well as below it.

Examples of RREF

A 2×3 example:

$$\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{A } 4 \times 6 \text{ example:} \quad \begin{bmatrix} 0 & 1 & * & 0 & * & 0 \\ 0 & 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2.3. Two Key Theorems

Theorem 2.1 (Uniqueness of RREF)**.** Any matrix can be transformed into a unique reduced row echelon form using elementary row operations.

Theorem 2.2 (Invariance)**.** Elementary row operations never change the solution set of the underlying linear system.

2.4. Pivots and Free Variables

Once a matrix is in echelon form, the following terms describe its structure:

- **Pivot** $\rightarrow$ The non-zero leading entry of a row.
- **Pivot Column** $\rightarrow$ A column that contains a pivot (non-zero leading entry).
- **Pivot Variable** $\rightarrow$ The variable corresponding to a pivot column.
- **Free Column** $\rightarrow$ A non-pivot column in the coefficient matrix (excluding the augmented column).
- **Free Variable** $\rightarrow$ The variable corresponding to a non-pivot column.

Remark 2.1*.* Note for augmented columns: the final column of an augmented matrix (the right-hand constants) is **NOT** a free column — it does not belong to the coefficient matrix or correspond to any variable.

2.5. The Reduction Method: Two Passes

Step 1 — Forward Pass (top-left to bottom-right)

Work from top-left to bottom-right, using elementary row operations to bring the matrix into Echelon Form (REF).

Step 2 — Backward Pass (bottom-right to top-left)

Work from bottom-right to top-left to bring the matrix into Reduced Row Echelon Form (RREF): work backward to create zeros above each pivot and scale pivots to 1.

2.6. Worked Example: Full Reduction to RREF

Example 2.1. Put the following matrix into reduced row echelon form, then find the general solution.

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{array}\right]$$

Part (a) — Row Operations to RREF

Forward Pass (to Echelon Form)

Swap R_1 and R_3 $(R_1 \leftrightarrow R_3)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{array}\right]$$

Eliminate R_2's leading entry $(-2R_1 + R_2 \to R_2)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{array}\right]$$

Swap R_2 and R_3 $(R_2 \leftrightarrow R_3)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \end{array}\right]$$

Eliminate the entry below the pivot $(R_3 - 2R_2 \to R_3)$ — this is the Echelon Form:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Backward Pass (to RREF)

Clear the entries above the pivot in Column 5 $(R_2 + R_3 \to R_2)$ and $(R_1 - 2R_3 \to R_1)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 0 & -5 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Clear the entry above the pivot in Column 3 $(R_1 - R_2 \to R_1)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

2.7. Reading Off the General Solution

Continuing Example 2.1, from the RREF augmented matrix:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Identifying the Variables

- **Pivot Columns:** $1, 3, 5$ $\to$ **Pivot Variables:** x_1, x_3, x_5
- **Free Columns:** $2, 4$ $\to$ **Free Variables:** x_2, x_4

System Equations

$$\begin{aligned} x_1 - x_2 - 5x_4 &= -9 \\ x_3 + 4x_4 &= 4 \\ x_5 &= 3 \end{aligned}$$

General Solution

$$\begin{aligned} x_1 &= x_2 + 5x_4 - 9 \\ x_2 &= \textbf{free} \\ x_3 &= -4x_4 + 4 \\ x_4 &= \textbf{free} \\ x_5 &= 3 \end{aligned}$$

2.8. Consistent vs. Inconsistent Systems

- A **consistent system** is a linear system that has at least one solution (unique or infinitely many).
- An **inconsistent system** is a linear system that has no solutions.

Remark 2.2. When reducing an augmented matrix $[\,A \mid \vec{b}\,]$, exactly one of three cases will apply — covered in the next three sections.

2.9. Case 1: No Solution

Condition

- The last (constant) column contains a pivot.
- **Equivalently:** a row of the form $[\,0\ 0\ \cdots\ 0 \mid \blacksquare\,]$ exists, with $\blacksquare \neq 0$.

Reason: such a row translates to the equation $0 = \blacksquare$, which is logically impossible.

Example

$$\left[\begin{array}{ccc|c} 1 & 0 & 7 & -12 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 5 \end{array}\right] \quad \Longrightarrow \quad \begin{aligned} x_1 + 7x_3 &= -12 \\ x_3 &= 7 \\ 0 &= 5 \ \leftarrow \ \textbf{Impossible!} \end{aligned}$$

Conclusion: No Solution (Inconsistent)

2.10. Case 2: A Unique Solution

Condition

- The last column is **NOT** a pivot column, **AND**
- There are **NO** free variables — every coefficient column has a pivot.

Example

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 5 & 15 \end{array}\right]$$

Method 1 — Back-Substitution from Echelon Form

Row 2: $5x_2 = 15 \Rightarrow x_2 = 3$; Row 1: $x_1 - x_2 = 4 \Rightarrow x_1 - 3 = 4 \Rightarrow x_1 = 7.$

Method 2 — Reduce Fully to RREF First

Divide Row 2 by 5 $(R_2/5 \to R_2)$:

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 1 & 3 \end{array}\right]$$

Add R_2 to R_1 $(R_1 + R_2 \to R_1)$:

$$\left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & 3 \end{array}\right]$$

Unique Solution: $x_1 = 7$, $x_2 = 3$

2.11. Case 3: Infinitely Many Solutions

Condition

- The last column is **NOT** a pivot column, **AND**
- At least one free variable exists — at least one coefficient column has no pivot.

Example — RREF Matrix

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

1. Identify Variables

- Pivot Columns: $1, 3, 5 \to$ Pivot Variables: x_1, x_3, x_5
- Free Columns: $2, 4 \to$ Free Variables: x_2, x_4

2. System Equations

$$\begin{aligned} x_1 - x_2 - 5x_4 &= -9 \\ x_3 + 4x_4 &= 4 \\ x_5 &= 3 \end{aligned}$$

3. General Solution (isolate the pivot variables)

$$\begin{aligned} x_1 &= x_2 + 5x_4 - 9 \\ x_2 &= \textbf{free} \\ x_3 &= -4x_4 + 4 \\ x_4 &= \textbf{free} \\ x_5 &= 3 \end{aligned}$$

Particular Solution

Setting the free variables $x_2 = 0$ and $x_4 = 0$ gives one specific solution vector:

$$(x_1,\, x_2,\, x_3,\, x_4,\, x_5) = (-9,\, 0,\, 4,\, 0,\, 3)$$

2.12. A Quick Decision Table

Table 2.1: Deciding the number of solutions from the row-reduced augmented matrix.

Last column has a pivot?	**Free variables?**	**System status**	**Number of solutions**
Yes $[0 \;\cdots\; 0 \mid \blacksquare]$	N/A	Inconsistent	0 Solutions
No	No	Consistent	1 Unique Solution
No	Yes	Consistent	Infinitely Many

2.13. Summary

- Echelon Form (REF) requires non-zero rows on top and staircase-pattern leading entries.
- RREF additionally requires all pivots to equal 1, with zeros above AND below each pivot.
- Every matrix has a unique RREF, reachable through elementary row operations.
- Pivot columns fix pivot variables; non-pivot columns give free variables.
- Reduction proceeds in two passes: forward to REF, then backward to RREF.
- A pivot in the constants column always signals an inconsistent (no-solution) system.
- No free variables and a consistent system → exactly one unique solution.
- At least one free variable and a consistent system → infinitely many solutions.

The next chapter turns from matrices to vectors: the spaces $\mathbb{R}^n$, linear combinations, and span.

Chapter 3

Vectors, Linear Combinations and Span

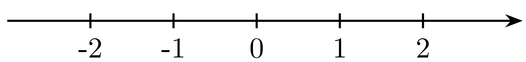
3.1. Goals

- Learn what vectors are, and what $\mathbb{R}^n$ represents.
- Define a linear combination and the span of a set of vectors.
- Understand how to determine whether a vector $\vec{b}$ can be written as a linear combination of a set of vectors — both computationally and conceptually.

3.2. The Spaces $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$

$\mathbb{R}$ — Real Numbers (1-Dimensional)

Represents the real number line:



$\mathbb{R}^2$ — The Plane

$$\mathbb{R}^2 = \{ \text{ all } (x, y) \text{ such that } x, y \in \mathbb{R} \ \} \qquad \text{(set of ordered pairs)}$$

$\mathbb{R}^3$ — 3D Space

$$\mathbb{R}^3 = \{ \text{ all } (x, y, z) \text{ such that } x, y, z \in \mathbb{R} \ \} \qquad \text{(set of ordered triples)}$$

General Definition — $\mathbb{R}^n$

$$\mathbb{R}^n := \{ \text{ all possible set of ordered } n\text{-tuples } (x_1, x_2, \ldots, x_n) \text{ of reals } \}$$

where ":=" means "defined to be equal to."

Remark 3.1. Elements of $\mathbb{R}^n$ are usually written as column vectors:

$$\mathbb{R}^n = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \text{where } x_1, x_2, \ldots, x_n \in \mathbb{R}.$$

3.3. Vector Notation and the Zero Vector

Definition 3.1 (General vector)**.** We write a general vector in $\mathbb{R}^n$ as

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

(The arrow above x represents that it is a vector.)

Definition 3.2 (Zero vector)**.**

$$\vec{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^n.$$

Remark 3.2. The individual values inside a vector are called *entries* of the vector.

3.4. Vector Operations

Let $\vec{x} = [x_1, \dots, x_n]$, $\vec{y} = [y_1, \dots, y_n] \in \mathbb{R}^n$, and let $c \in \mathbb{R}$ be a scalar (constant).

Definition 3.3 (Vector addition)**.**

$$\vec{x} + \vec{y} := \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Definition 3.4 (Vector subtraction)**.**

$$\vec{x} - \vec{y} := \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}$$

Definition 3.5 (Scalar multiplication (scaling))**.**

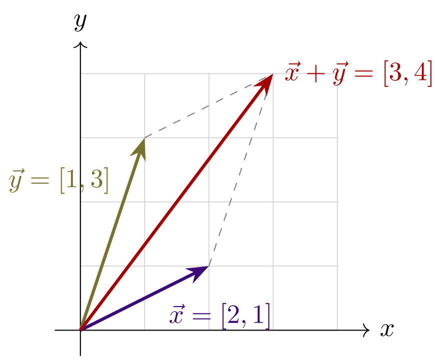
$$c\,\vec{x} = \begin{bmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{bmatrix} \qquad \text{(Scalar)} \cdot \text{(Vector)} = \text{Vector}$$

3.5. The Geometry of $\mathbb{R}^2$

Convention: Always start vectors at the origin (0, 0).

Example 3.1 (Vector addition and the parallelogram rule)**.** Let $\vec{x} = [2,\ 1]$ and $\vec{y} = [1,\ 3] \in \mathbb{R}^2$:

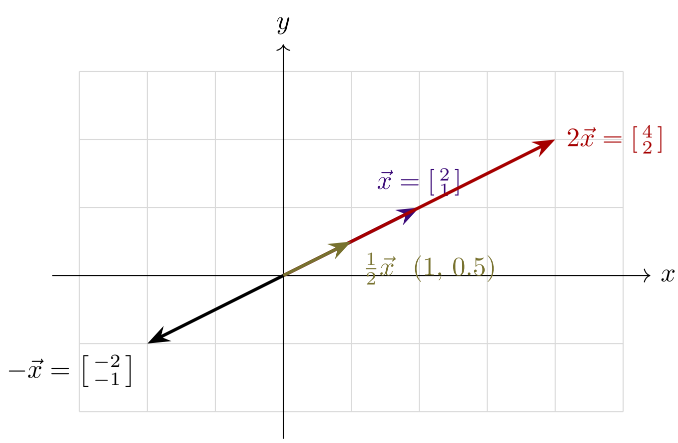
$$\vec{x} + \vec{y} = [2 + 1,\ 1 + 3] = [3,\ 4].$$



Picture in $\mathbb{R}^2$: draw $\vec{x}$, $\vec{y}$, and $\vec{x} + \vec{y}$ in the xy-plane — parallelogram addition.

Example 3.2 (Scalar scaling)**.** Let $\vec{x} = [2,\ 1]$. We sketch $\vec{x}$, $2\vec{x}$, $\frac{1}{2}\vec{x}$, and $-\vec{x}$:

- $2\vec{x} = [4,\ 2]$
- $\frac{1}{2}\vec{x} = [1,\ 0.5]$
- $-\vec{x} = [-2,\ -1]$



3.6. Arithmetic Properties of Vectors

Vectors in $\mathbb{R}^n$ satisfy fundamental algebraic properties for any vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$:

Associativity: $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ **Additive identity:** $\vec{0} + \vec{v} = \vec{v}$

Zero scalar: $0 \cdot \vec{v} = \vec{0}$ **Distributivity:** $c(\vec{u} + \vec{v}) = c\,\vec{u} + c\,\vec{v}$

3.7. Linear Combinations

Definition 3.6 (Linear combination)**.** Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$ be vectors in $\mathbb{R}^n$. Let $c_1, c_2, \ldots, c_k \in \mathbb{R}$ be scalars (called *weights*). Then the expression

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$$

is called a **linear combination** of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$.

Remark 3.3. A linear combination of vectors results in another vector.

3.8. Span

Definition 3.7 (Span)**.** The **span** of a set of vectors is the set of all possible linear combinations of those vectors:

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} = \text{all linear combinations of } \vec{v}_1, \ldots, \vec{v}_k.$$

Conceptually:

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} = \text{everywhere we can get to in } \mathbb{R}^n \text{ only traveling in the directions of } \vec{v}_1, \ldots, \vec{v}_k.$$

Key Fact. The zero vector $\vec{0}$ is always in $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$. Why? By choosing all weights to be 0:

$$0\vec{v}_1 + 0\vec{v}_2 + \cdots + 0\vec{v}_k = \vec{0}.$$

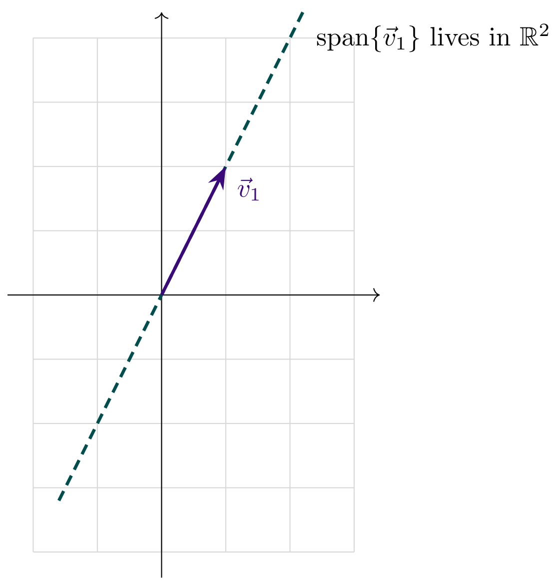
3.9. Geometric Examples of Span

(a) A Single Vector in $\mathbb{R}^2$

Example 3.3. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Then

$$\text{Span}\{\vec{v}_1\} = \{\, c\,\vec{v}_1 : c \in \mathbb{R}\,\},$$

which is a line through the origin.



(b) Two Linearly Independent Vectors in $\mathbb{R}^3$

Example 3.4. Let $\vec{v}_1 = [1,\ 0,\ 0]$ and $\vec{v}_2 = [0,\ 1,\ 0]$ in $\mathbb{R}^3$:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1[1, 0, 0] + c_2[0, 1, 0] = [\,c_1,\ c_2,\ 0\,].$$

Geometric interpretation: the xy-plane in 3D-space ($\mathbb{R}^3$).

Remark 3.4 (Important). $\text{Span}\{\vec{v}_1, \vec{v}_2\} \neq \mathbb{R}^2$, because the vectors in this span have 3 components and live in $\mathbb{R}^3$, whereas elements of $\mathbb{R}^2$ only have 2 components.

(c) Two Parallel (Dependent) Vectors in $\mathbb{R}^3$

Example 3.5. Let $\vec{v}_1 = [1,\ 3,\ -4]$ and $\vec{v}_2 = \left[\frac{1}{4},\ \frac{3}{4},\ -1\right]$ in $\mathbb{R}^3$. Observe that $\vec{v}_1 = 4\vec{v}_2$. Substituting into a general linear combination gives:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1(4\vec{v}_2) + c_2\vec{v}_2 = (4c_1 + c_2)\,\vec{v}_2.$$

Since $(4c_1 + c_2)$ is just a scalar constant:

$$\mathrm{Span}\{\vec{v}_1,\ \vec{v}_2\} = \mathrm{Span}\{\vec{v}_2\} = \text{the line through the origin and } \left[\tfrac{1}{4},\ \tfrac{3}{4},\ -1\right] \text{ in } \mathbb{R}^3.$$

3.10. Matrix Representation of a Linear System

Suppose a linear system has m equations and n variables. Its augmented matrix is expressed as:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] = \left[\begin{array}{cccc|c} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{array}\right] = [\,A \mid \vec{b}\,]$$

- $A = [\,\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\,]$ is the $m \times n$ **coefficient matrix**.
- Each column $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$ is a column vector in $\mathbb{R}^m$.
- $\vec{b} = [b_1, b_2, \ldots, b_m]$ is the right-hand side constant vector in $\mathbb{R}^m$.

3.11. Three Equivalent Descriptions of a Solution

Let $\vec{x} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^n$ be a candidate solution vector for $[\,A \mid \vec{b}\,]$. The following statements are equivalent.

1. System of Equations Representation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

2. Vector Equation Representation

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \qquad \Longleftrightarrow \qquad x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

3. Linear Combination / Span Interpretation

$\vec{b}$ is a linear combination of the column vectors $\vec{a}_1, \ldots, \vec{a}_n$.

Equivalently: $\vec{b} \in \text{Span}\{\vec{a}_1, \ \dots, \ \vec{a}_n\}$

3.12. The Matrix–Vector Product

Definition 3.8 (Matrix–vector product)**.** If $A = [\,\vec{a}_1 \ \vec{a}_2 \ \cdots \ \vec{a}_n\,]$ is an $m \times n$ matrix (with m rows and n columns, where each $\vec{a}_i \in \mathbb{R}^m$) and $\vec{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$, then the product $A\vec{x}$ is defined as:

$$A\vec{x} := x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

Remark 3.5. This product $A\vec{x}$ represents a linear combination of the columns of A with weights given by the entries of $\vec{x}$, resulting in a vector in $\mathbb{R}^m$.

3.13. Existence of Solutions

Theorem 3.1 (Existence of solutions)**.** The following statements are equivalent:

The linear system $[\,A \mid \vec{b}\,]$ has a solution

- $\iff$ $\vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ (same as saying $\vec{b} \in \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$);
- $\iff$ $A\vec{x} = \vec{b}$ for some $\vec{x} \in \mathbb{R}^n$;
- $\iff$ the echelon form of $[\,A \mid \vec{b}\,]$ has NO row of the form $[\,0 \ \cdots \ 0 \mid \blacksquare\,]$ with $\blacksquare \neq 0$.

3.14. Worked Example: Testing Span Membership

Example 3.6. Is

$$[1,\, 2,\, 3] \in \text{Span}\{\,[-1, 1, 2],\ [2, 1, 0],\ [1, 2, 2]\,\}\,?$$

(i.e., is $[1, 2, 3]$ a linear combination of those vectors?)

Check whether the linear system given by the augmented matrix is consistent. Starting matrix:

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & 2 & 3 \end{array}\right]$$

Apply row operations $R_1 + R_2 \to R_2,\ 2R_1 + R_3 \to R_3$:

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 4 & 4 & 5 \end{array}\right]$$

Simplify rows: $(-1)R_1 \to R_1,\ (1/3)R_2 \to R_2$:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 4 & 4 & 5 \end{array}\right]$$

Eliminate Row 3: $-4R_2 + R_3 \to R_3$:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The row $[\,0\ 0\ 0 \mid 1\,]$ indicates that the system is inconsistent.

$[1,\ 2,\ 3]$ **is NOT in** $\text{Span}\{\,[-1,1,2],\ [2,1,0],\ [1,2,2]\,\}$

3.15. Worked Example: Finding Weights in a Span

Example 3.7. Is

$$\vec{b} = [-9,\ 3,\ 10] \in \text{Span}\{\,[-1,1,2],\ [2,1,0]\,\}\,?$$

(If yes, find weights x_1, x_2 such that $\vec{b} = x_1\vec{v}_1 + x_2\vec{v}_2$.)

Set up the augmented matrix:

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 1 & 1 & 3 \\ 2 & 0 & 10 \end{array}\right]$$

Row operations to Echelon Form: $R_1 + R_2 \to R_2,\ \ 2R_1 + R_3 \to R_3$:

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 0 & 3 & -6 \\ 0 & 4 & -8 \end{array}\right]$$

Rescale rows: $R_1(-1) \to R_1,\ \ R_2(1/3) \to R_2,\ \ R_3(1/4) \to R_3$:

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{array}\right]$$

Eliminate Row 3: $R_2 - R_3 \to R_3$, giving the Echelon Form:

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$$

Observation: the Echelon Form has no row of the form $[\,0 \ldots 0 \mid \blacksquare\,]$ where $\blacksquare \neq 0$ — thus a solution exists.

Back substitution / RREF: $2R_2 + R_1 \to R_1$:

$$\left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \qquad \Longrightarrow \qquad x_1 = 5, \quad x_2 = -2.$$

Thus:

$$[-9,\ 3,\ 10] \ = \ 5 \cdot [-1,\ 1,\ 2] \ - \ 2 \cdot [2,\ 1,\ 0]$$

3.16. Spanning the Entire Space

Theorem 3.2 (Spanning $\mathbb{R}^m$)**.** Let A be an $m \times n$ matrix with columns $\vec{a}_1, \dots, \vec{a}_n$. The following statements are equivalent ($\Longleftrightarrow$):

1. The linear system $[\, A \mid \vec{b}\,]$ has a solution for every $\vec{b} \in \mathbb{R}^m$.
2. $\operatorname{Span}\{\vec{a}_1, \dots, \vec{a}_n\} = \mathbb{R}^m$.
3. The reduced matrix never has a row of the form $[\, 0 \dots 0 \mid \blacksquare\,]$ (where $\blacksquare \neq 0$).
4. The coefficient matrix A has a pivot in every row.

3.17. Worked Example: Testing a Full-Space Span

Example 3.8. Do the vectors $[1, 2, 3]$, $[1, 0, 1]$, $[2, -1, 1]$ span all of $\mathbb{R}^3$?

Form the matrix A and check whether there is a pivot in each row:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Eliminate the lower entries in Column 1: $-2R_1 + R_2 \to R_2$, $-3R_1 + R_3 \to R_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & -2 & -5 \end{bmatrix}$$

Eliminate Row 3: $R_3 - R_2 \to R_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Since row 3 contains all zeros, not every row has a pivot (there are only 2 pivots, but 3 rows are required).

The columns do NOT span $\mathbb{R}^3$.

3.18. Summary

- $\mathbb{R}^n$ is the set of ordered n-tuples, usually written as column vectors.
- Vectors can be added, subtracted, and scaled entrywise.
- Vector addition follows the parallelogram rule geometrically.
- A linear combination $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$ always produces another vector.

- Span$\{\vec{v}_1, \ldots, \vec{v}_k\}$ is the set of ALL linear combinations — and always contains $\vec{0}$.
- A linear system $[\, A \mid \vec{b}\,]$ has a solution $\iff \vec{b} \in \text{Span}\{\text{columns of } A\} \iff A\vec{x} = \vec{b}$ for some $\vec{x}$.
- A pivot-free row in the constants column of the echelon form guarantees a solution exists.
- Columns of A span the ENTIRE space $\mathbb{R}^m$ only when A has a pivot in every row.

The next chapter studies the matrix equation $A\vec{x} = \vec{b}$ in its own right.

Chapter 4

The Matrix Equation $A\vec{x} = \vec{b}$

4.1. Goals

- **Goal 1:** Learn how to compute $A\vec{x}$ in two ways.
- **Goal 2:** Understand how to write a linear system in **vector form** and **matrix form**.

4.2. Defining Matrix–Vector Multiplication

Definition 4.1 (Matrix–vector multiplication)**.** Let $A = [\,\vec{a}_1\ \vec{a}_2\ \cdots\ \vec{a}_n\,]$ be an $m \times n$ matrix (m rows, n columns), where each column vector $\vec{a}_i \in \mathbb{R}^m$. Let

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

Then the product $A\vec{x}$ is defined as a **linear combination of the columns of** A:

$$A\vec{x} \;=\; \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \;:=\; x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

4.3. Computing $A\vec{x}$: Two Methods

Example 4.1. Compute the following, if defined.

Part (a)

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix}$$

Method 1: Linear Combination of Columns.

$$= 5\begin{bmatrix} 0 \\ 2 \end{bmatrix} - 1\begin{bmatrix} -3 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

Method 2: Row-Vector Rule (Dot Product Rule).

$$\begin{bmatrix} 5(0) + (-1)(-3) + 2(5) \\ 5(2) + (-1)(1) + 2(4) \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

Part (b)

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

- **Result:** Undefined.
- **Reason:** The number of columns of A (3) does not match the number of entries in the vector $\vec{x}$ (2).

Part (c)

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 5 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 7 \end{bmatrix}$$

Using the Row-Vector Rule:

$$= \begin{bmatrix} 3(1) + 0(-2) + 7(1) \\ 3(0) + 0(0) + 7(5) \\ 3(0) + 0(3) + 7(-1) \end{bmatrix} = \begin{bmatrix} \mathbf{10} \\ \mathbf{35} \\ \mathbf{-7} \end{bmatrix}$$

4.4. Vector Form and Matrix Form of a System

Example 4.2. Given the system of linear equations:

$$\begin{aligned} 6x_1 - x_2 &= 9 \\ 4x_1 + 3x_2 &= -5 \\ x_1 + 7x_2 &= -20 \end{aligned}$$

Vector form:

$$x_1 \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

Matrix form $(A\vec{x} = \vec{b})$:

$$\begin{bmatrix} 6 & -1 \\ 4 & 3 \\ 1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

4.5. Solving the Matrix Equation

Example 4.3 (continuation of Example 4.2)**.** Solve $A\vec{x} = \vec{b}$ using row reduction.

1. **Augmented matrix:**

$$\left[\begin{array}{cc|c} 6 & -1 & 9 \\ 4 & 3 & -5 \\ 1 & 7 & -20 \end{array}\right]$$

2. **Swap rows** $(R_1 \leftrightarrow R_3)$:

$$\Rightarrow \quad \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 4 & 3 & -5 \\ 6 & -1 & 9 \end{array}\right]$$

3. **Eliminate lower column 1 entries:** $-4R_1 + R_2 \to R_2$; $-6R_1 + R_3 \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & -25 & 75 \\ 0 & -43 & 129 \end{array}\right]$$

4. **Rescale rows:** $R_2\left(-\frac{1}{25}\right) \to R_2$; $R_3\left(-\frac{1}{43}\right) \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{array}\right]$$

5. **Eliminate Row 3:** $R_3 - R_2 \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

6. **Reduced Row Echelon Form:** $-7R_2 + R_1 \to R_1$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

Final Solution: $x_1 = 1$**,** $x_2 = -3$**, i.e.** $\vec{x} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

The final chapter examines the structure of solution sets: homogeneous and nonhomogeneous systems, and the geometry behind them.

Chapter 5

Solution Sets of Linear Systems

5.1. Goals

- **Goal 1:** Learn how to write the general solution to a linear system in **parametric vector form**.
- **Goal 2:** Understand geometrically the difference in the solution sets of $A\vec{x} = \vec{0}$ and $A\vec{x} = \vec{b}$ when $\vec{b} \neq \vec{0}$.

5.2. Homogeneous Linear Systems

Definition 5.1 (Homogeneous system)**.** A linear system is called **homogeneous** if it can be written in the form

$$A\vec{x} = \vec{0},$$

where A is an $m \times n$ matrix and $\vec{0}$ is the zero vector in $\mathbb{R}^m$.

In terms of the columns of A:

$$A\vec{x} = \vec{0} \iff x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{0}.$$

Remark 5.1. The vector $\vec{x} = \vec{0}$ (i.e., $x_1 = x_2 = \cdots = x_n = 0$) is **always** a solution to a homogeneous system. This is called the **trivial solution**. Nonzero solutions, if they exist, are called **nontrivial solutions**.

5.3. Worked Example: A Homogeneous System

Example 5.1. Find the solution set of the following system in **parametric vector form**:

$$\begin{aligned} x_3 + 4x_4 - x_5 &= 0 \\ 2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 &= 0 \\ x_1 - x_2 + x_3 - x_4 + 2x_5 &= 0 \end{aligned}$$

1. **Augmented matrix & row reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 0 \\ 2 & -2 & 4 & 6 & 3 & 0 \\ 1 & -1 & 1 & -1 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & 0 \\ 0 & 0 & 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

2. **Identify variables:** pivot variables x_1, x_3, x_5; free variables x_2, x_4.
3. **Express pivot variables in terms of the free variables:**

$$\begin{aligned} x_1 - x_2 - 5x_4 = 0 &\implies x_1 = x_2 + 5x_4 \\ x_3 + 4x_4 = 0 &\implies x_3 = -4x_4 \\ x_5 = 0 &\implies x_5 = 0 \end{aligned}$$

Writing every variable explicitly:

$$\begin{aligned} &x_1 = x_2 + 5x_4 && x_2 = \textit{free} && x_3 = -4x_4 \\ &x_4 = \textit{free} && x_5 = 0 \end{aligned}$$

4. **General solution in parametric vector form:**

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 + 5x_4 \\ x_2 \\ -4x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

Parametric vector form: $\vec{x} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$

5.4. Nonhomogeneous Linear Systems

Definition 5.2 (Nonhomogeneous system)**.** A linear system is called **nonhomogeneous** if it can be written in the form

$$A\vec{x} = \vec{b} \quad \text{where} \quad \vec{b} \neq \vec{0}.$$

5.5. Worked Example: A Nonhomogeneous System

Example 5.2. Find the solution set of the following system in **parametric vector form**:

$$\begin{aligned} x_3 + 4x_4 - x_5 &= 1 \\ 2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 &= 7 \\ x_1 - x_2 + x_3 - x_4 + 2x_5 &= 1 \end{aligned}$$

1. **Augmented matrix & row reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

2. **Express pivot variables in terms of the free variables:**

$$\begin{aligned} x_1 - x_2 - 5x_4 = -9 &\implies x_1 = -9 + x_2 + 5x_4 \\ x_3 + 4x_4 = 4 &\implies x_3 = 4 - 4x_4 \\ x_5 = 3 &\implies x_5 = 3 \end{aligned}$$

Writing every variable explicitly:

$$\begin{aligned} &x_1 = -9 + x_2 + 5x_4 && x_2 = \textit{free} && x_3 = 4 - 4x_4 \\ &x_4 = \textit{free} && x_5 = 3 \end{aligned}$$

3. **General solution in parametric vector form:**

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -9 + x_2 + 5x_4 \\ x_2 \\ 4 - 4x_4 \\ x_4 \\ 3 \end{bmatrix} = \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

Parametric vector form: $\vec{x} = \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$

5.6. The Solution Structure Theorem

Notice how the solution vector $\vec{x}$ to $A\vec{x} = \vec{b}$ in Example 5.2 splits into two distinct components:

$$\vec{x} = \underbrace{\begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix}}_{\vec{v}_p} + \underbrace{x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}}_{\vec{v}_h}$$

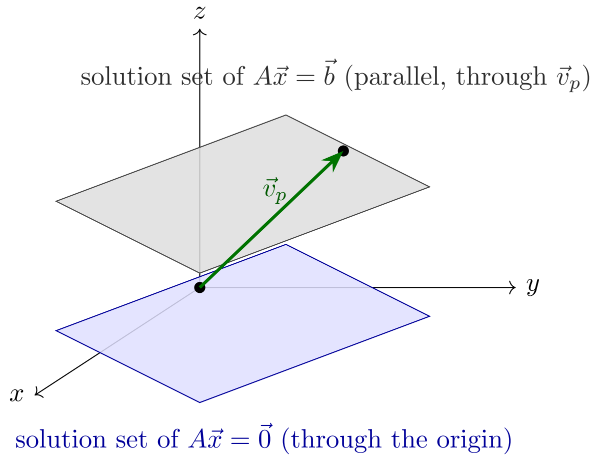
- $\vec{v}_p$: a **particular solution** to $A\vec{x} = \vec{b}$ (satisfies $A\vec{v}_p = \vec{b}$).
- $\vec{v}_h$: the **homogeneous solution** (the general solution to $A\vec{x} = \vec{0}$).

Theorem 5.1 (Solution structure)**.** Suppose the equation $A\vec{x} = \vec{b}$ is consistent for a given vector $\vec{b}$, and let $\vec{v}_p$ be a particular solution. Then the solution set of $A\vec{x} = \vec{b}$ (where $\vec{b} \neq \vec{0}$) is the set of all vectors of the form

$$\vec{x} = \vec{v}_p + \vec{v}_h,$$

where $\vec{v}_h$ is any solution of the corresponding homogeneous system $A\vec{x} = \vec{0}$.

5.7. The Geometric Picture in $\mathbb{R}^3$



- The solution to $A\vec{x} = \vec{b}$ is **parallel** to the solution to $A\vec{x} = \vec{0}$ and goes through $\vec{v}_p$.
- The solution to $A\vec{x} = \vec{0}$ goes through the origin, since $\vec{x} = \vec{0}$ is a solution to $A\vec{x} = \vec{0}$.

The geometric picture in $\mathbb{R}^3$ for a consistent linear system $A\vec{x} = \vec{b}$ is an **affine plane**: a translation of the solution subspace of the homogeneous system $A\vec{x} = \vec{0}$ (a plane through the origin) by any particular solution $\vec{v}_p$ of $A\vec{x} = \vec{b}$.

Remark 5.2. If $A\vec{x} = \vec{b}$ is inconsistent, there is no geometric picture (no intersection). If it has a unique solution, the picture reduces to a single point.

Summary of the picture: The solution set of $A\vec{x} = \vec{b}$ is **parallel** to the solution set of $A\vec{x} = \vec{0}$, and passes through $\vec{v}_p$. The solution set of $A\vec{x} = \vec{0}$ passes through the origin, since $\vec{x} = \vec{0}$ is always a solution to $A\vec{x} = \vec{0}$.

Looking ahead. *The natural next topic in the series is* linear independence.